Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/13/2019
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	VNSCOR: Mechanisms of Radiation-Induced Changes in Sustained Attention and Social Processing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	catherine.davis-takacs@usuhs.edu	**Fax:**	FY 301-400-4023
PI Organization Type:	NON-PROFIT	**Phone:**	301-400-4596
Organization Name:	Henry M. Jackson Foundation		
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City:	Bethesda	**State:**	MD
Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	04/15/2018	**End Date:**	10/01/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2020 (officially 9/30/2020) due to PI move to Henry M. Jackson Foundation in fall 2020 (Ed., 1/18/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University) Robinson, Siobhan Ph.D. (Hamilton College)		
Grant/Contract No.:	80NSSC18K1080		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note Jan 2019: See also project, "VNSCOR: Responses of the Nervous System to Chronic, Low Dose Charged Particle Irradiation" (PI: Greg Nelson)] NELSON/DAVIS VIRTUAL NASA Specialized Center of Research (NSCOR): The project is organized as 5 large experimental campaigns to quantify responses for an interrelated set of central nervous system (CNS) outcome measures in mice to acute and protracted exposures to protons at a dose of 0.5 Gy and sham controls; acute and protracted exposures to 0.25 and 0.5 Gy of charged particles; and acute and protracted exposures to 0.5 and 1.5 Gy of gamma rays. This research builds on previous studies that demonstrated that proton and HZE (high charge energy) exposures result in individual differences in deficits in sustained attention, but more general deficits in recognition memory. This current project is combined with "Responses to the Nervous System to Chronic, Low Dosed Charged Particle Irradiation" (PI: Nelson) in order to explore if these effects are LET (linear energy transfer)-dependent for 16O ions, add a relatively understudied, but important, ion (4He), and examine CNS effects in whole animals following fractionated exposures, and the interaction of other space flight factors (e.g., sleep fragmentation). Revised Specific Aims: Aim 1a: Effects of protracted exposure to five-ion GCR sim (1H, 4He, 28Si, 16O, 56Fe) Aim 1b: Effects of protracted exposure to GCR sim without protons (4He, 28Si, 16O, 56Fe) Aim 1c: Effects of protracted exposure to protons only Aim 1d: Acute exposure to 4He (250 MeV/n) Aim 2a: Neuronal activation and molecular markers following radiation Aim 2b: Chemogenetic silencing of mPFC subregions Specific Aims: 1) Determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance. (This aim has been modified in order to integrate with Nelson project); 2) Determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) Determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure (This aim has been modified in order to integrate with Nelson project); 4) Examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. ORIGINAL PROPOSAL DESCRIPTION: Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to 1) determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance; 2) determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure; and 4) examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in sustained attention and 2) shown that acute, single ion exposures alter social motivation and social odor recognition memory. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing various behavioral, physiological, and neurobiological markers of radiation exposure at early and later time points post-exposure. Groups of animals will be trained on a rodent version of the human psychomotor vigilance test, exposed to radiation, and then tested for both social processing and sustained attention deficits following exposure; subsets of rats will be implanted with biotelemetry devices to measure fluctuations in spontaneous locomotor activity and body temperatures following radiation exposure and changes in sleep or circadian disruptions. Individual variations in these behavioral, physiological, or neurobiological responses following radiation will be assessed. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to a pharmacogenetic technique to determine the severity of radiation-induced neurobehavioral deficits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the current project will be used to understand how radiation exposure affects the central nervous system to induced deficits in neurobehavioral function.
Task Progress:	This is the first year of this project and as such, progress is limited, given that NASA Space Radiation Laboratory (NSRL) campaigns have just been completed and/or we are preparing for future campaigns. During Year 1 of funding, we initiated a study examining the effects of acute helium (4He) exposure on sustained attention and social odor recognition in rats. To date, we have completed the first three time points of this study (7 days, 30 days, and 90 days following radiation exposure). Behavioral and immunohistochemical data are currently being acquired and analyzed. Also during Year 1, we began preparing for two additional NSRL campaigns where we will expose subjects to the five-ion galactic cosmic radiation (GCR) simulation using either a protracted exposure (delivered over 4 weeks) or an acute exposure, and will compare that to the same five-ion GCR simulation delivered acutely (in on day).
Bibliography Type:	Description: (Last Updated: 06/30/2026)

Abstracts for Journals and Proceedings	Granata LM, Robinson S, Hienz RD, Davis CM. "The Medial Prefrontal Cortex is Involved in Social Odor Recognition Memory." Annual Meeting of the Federation of American Societies for Experimental Biology. Experimental Biology 2018, San Diego, CA, April 21-25, 2018. FASEB Journal. 2018 Apr;32(Suppl 1):551.1. , Apr-2018
Articles in Other Journals or Periodicals	Davis CM, Rabin BM. "Space radiation-induced cognitive deficits following head only, whole body, or partial body exposures." The Health Risks of Extraterrestrial Environments (THREE). https://three.jsc.nasa.gov (in press, as of February 2019). , Feb-2019
Articles in Peer-reviewed Journals	Robinson S, Granata L, Hienz RD, Davis CM. "Temporary inactivation of the medial prefrontal cortex impairs the formation, but not the retrieval of social odor recognition memory in rats." Neurobiol Learn Mem. 2019 May 3;161:115-21. Epub 2019 Apr 3. https://doi.org/10.1016/j.nlm.2019.04.003 ; PubMed PMID: 30953696 , May-2019
Articles in Peer-reviewed Journals	Jones CB, Mange A, Granata L, Johnson B, Hienz RD, Davis CM. "Short and long-term changes in social odor recognition and plasma cytokine levels following oxygen (16O) ion radiation exposure." Int J Mol Sci. 2019 Jan 15;20(2): E339. https://doi.org/10.3390/ijms20020339 ; PubMed PMID: 30650610; PubMed Central PMCID: PMC6359552 , Jan-2019
Articles in Peer-reviewed Journals	Johnson D, Lawrence SE, Livingston EW, Hienz RD, Davis CM, Lau AG. "Modeling space radiation induced bone changes in rat femurs through finite element analysis." Conf Proc IEEE Eng Med Biol Soc. 2018 Jul;2018:1763-6. https://doi.org/10.1109/EMBC.2018.8512620 ; PubMed PMID: 30440736 , Jul-2018
Awards	Sfanos KS, Davis CM. "Seed Grant from Space@Hopkins to support the graduate stipend of Carli Jones, so she can collect fecal samples and GI tract samples from the irradiated rats supported by this NASA grant. Dates: 06/25/2018 - 06/24/2019." Jun-2018